

NCollection ERP: Project Master Context

[!WARNING] **SUPERSEDED (July 16, 2026)** — The task IDs (1.1 – 8.7), phase breakdown, and estimates in this document are **outdated**. The authoritative plan is [DELIVERABLE_1_SYSTEM_DESIGN.md](#) (v5.0) with [DELIVERABLE_2_TIMELINE_AND_TOOLING.md](#), [ARCHITECTURE_DATA_PLATFORM.md](#), and [ARCHITECTURE_SECURITY.md](#). This file is kept for historical context only — do not implement tasks from it.

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Purpose: This is the single, authoritative reference document for the entire NCollection ERP project. It combines the PRD, System Architecture, Developer Personas, and an exhaustive, highly detailed task breakdown for all 8 phases.

1. Executive Summary & Vision

NCollection ERP is a SaaS ERP Platform built on top of **Odoo 19 Community Edition**. The platform targets small-to-medium businesses (5–100 employees) in the UAE and GCC region. We are building a **SaaS platform layer** around Odoo (tenant management, subscription licensing, branding, UAE localization) so that subscribers land directly in an isolated ERP workspace containing only the modules they purchased.

2. Developer Personas & Roles

To ensure maximum parallelization, tasks are strictly assigned to three distinct developer roles:

[DEV-1] Backend & Infrastructure Lead

- **Specialty:** Python, PostgreSQL, Docker, CI/CD, SaaS Architecture, APIs.
- **Focus:** Tenant provisioning, DB routing, SaaS automation, API, AI integration.

[DEV-2] Odoo & Business Logic Specialist

- **Specialty:** Odoo Models (ORM), Workflows, XML Views, Access Rights, Core ERP.
- **Focus:** Module visibility, roles, OCA integrations, accounting logic, workflows.

[DEV-3] Frontend & Integration Specialist

- **Specialty:** OWL, QWeb, JavaScript, UI/UX, Mobile apps, Portals.
- **Focus:** Branding, customer dashboards, portals, mobile application, UI widgets.

3. Collaboration & Rules

1. **Never modify Odoo Core.** Use Custom Addons (`custom_addons/`) only.
2. **Issue Tracking:** Every task below maps to a GitHub Issue.
3. **Claude Code Workflow:** Assign task -> Branch -> Claude Code implementation -> Pull Request.
4. **Strict Dependencies:** Do not start a task if its listed dependency is incomplete.

4. Master Task Breakdown (All Phases)

Phase 1: Customer Workspace (Priority: Immediate)

Objective: Build the foundational SaaS experience where subscribers land directly in their isolated ERP workspace.

ID	Task Name	Description	Assigned	Dependencies
1.1	Docker & CI/CD Setup	Setup <code>docker-compose.yml</code> for PostgreSQL 16 and Odoo 19. Configure GitHub Actions to run <code>flake8</code> / <code>pylint-odoo</code> on PRs.	[DEV-1]	None (Start immediately)
1.2	Custom Addons Skeleton	Create the empty folder structure for <code>ncollection_core</code> , <code>ncollection_saas</code> , <code>ncollection_subscription</code> with <code>__manifest__.py</code> .	[DEV-1]	1.1
1.3	Tenant & Sub Models	Create <code>ncollection.tenant</code> and <code>ncollection.subscription</code> models with fields for database status, UUIDs, and limits.	[DEV-1]	1.2
1.4	DB Routing Engine	Implement logic using Odoo <code>--db-filter</code> or Nginx mapping to route subdomains (e.g. <code>client.ncollection.com</code>) to specific databases.	[DEV-1]	1.3
1.5	Customer Authentication	Override standard Odoo login flow to handle SaaS session management and "Remember Me" securely.	[DEV-1]	1.2
1.6	Module Visibility Engine	Write Python logic to intercept menu loading and hide Odoo apps based on the tenant's active subscription tier.	[DEV-2]	1.3
1.7	Tenant Roles Config	Define XML <code>res.groups</code> for standard SaaS roles: Owner, CEO, Manager, Sales, Warehouse, HR, Accountant.	[DEV-2]	1.2
1.8	App/Settings Stripping	Remove access to standard Odoo "Apps" and "Settings" menus for all standard tenant users to prevent tampering.	[DEV-2]	1.7

1.9	Web Client Branding	Complete <code>ncollection_branding</code> : replace Odoo logos, customize login page UI, update URL paths, and replace the About dialog.	[DEV-3]	1.2
1.10	Dynamic Tenant Branding	Inject CSS variables dynamically so each tenant database loads its own specific primary colors and logo upon login.	[DEV-3]	1.4, 1.9
1.11	Workspace Dashboard	Build the main Customer Dashboard widget (Sales, Receivables, quick actions) using the OWL framework.	[DEV-3]	1.2

Phase 2: SaaS Automation

Objective: Automate the business of selling ERP (provisioning, billing, renewals).

ID	Task Name	Description	Assigned	Dependencies
2.1	Provisioning Job Model	Create the <code>ncollection.provisioning.job</code> model to track DB creation queues and logs.	[DEV-1]	1.3
2.2	Auto-Provisioning Script	Write the Python script that executes Odoo CLI commands to spin up a new Postgres DB and install base modules automatically.	[DEV-1]	2.1
2.3	Backup Manager Script	Create a cron job/script to take <code>pg_dump</code> backups of all active tenant databases and upload to secure cloud storage.	[DEV-1]	2.2
2.4	Domain & SSL Manager	Automate Let's Encrypt SSL generation when a new subdomain is mapped to a tenant.	[DEV-1]	None
2.5	Sub Expiration Cron	Create a scheduled action to check subscription end dates and update tenant states to 'expired' or 'suspended'.	[DEV-1]	1.3
2.6	Billing Engine	Logic to automatically generate Odoo Invoices in the main Admin DB when a tenant purchases or renews a subscription.	[DEV-2]	1.3
2.7	SaaS Admin Dashboard	Build the internal views and action menus for NCollection staff to monitor tenant health and trigger manual provisions.	[DEV-2]	1.3
2.8	Public Checkout Flow	Build the frontend Odoo eCommerce/Website pages for users to select a plan and register their company.	[DEV-3]	None
2.9	Email Automation	Design HTML templates for welcome emails, renewal reminders, and suspension alerts, integrated with mail records.	[DEV-3]	None

Phase 3: ERP Enhancement

Objective: Improve core business modules and implement UAE localization.

ID	Task Name	Description	Assigned	Dependencies
3.1	OCA Financials Setup	Install and configure the <code>account_financial_report</code> OCA module. Ensure Odoo 19 compatibility.	[DEV-1]	1.1

3.2	Performance Tuning	Optimize <code>odoo.conf</code> (workers, memory limits) and Postgres configurations for heavy multi-tenant ERP usage.	[DEV-1]	1.1
3.3	UAE VAT & Currency	Configure base UAE localization: 5% VAT tax records and AED default currency setup via XML data.	[DEV-2]	1.1
3.4	UAE Chart of Accounts	Map and create the standard UAE Chart of Accounts using Odoo accounting templates.	[DEV-2]	3.1, 3.3
3.5	Workflow Enhancements	Customize standard CRM, Sales, and Purchase models to include multi-level approval workflows requested by clients.	[DEV-2]	None
3.6	Translation Audit	Export <code>.po</code> files and ensure 100% Arabic/English translation completeness across all <code>ncollection_*</code> modules.	[DEV-3]	None
3.7	UAE PDF Invoices	Design custom QWeb PDF templates for invoices that comply with UAE commercial and tax standards.	[DEV-3]	3.3
3.8	MIS Builder Templates	Complete the MIS Builder configurations for high-quality Balance Sheet and P&L financial reports.	[DEV-3]	3.1

Phase 4: Executive Dashboards

Objective: Provide high-level analytics for tenant executives.

ID	Task Name	Description	Assigned	Dependencies
4.1	Data Aggregation Engine	Write optimized SQL queries or Python methods to safely aggregate cross-module data (Sales + HR + Finance) quickly.	[DEV-1]	None
4.2	KPI Logic Models	Define backend models and computed fields to calculate specific metrics (e.g., Revenue Growth, Employee Turnover).	[DEV-2]	4.1
4.3	CEO Dashboard UI	Build a responsive OWL dashboard combining high-level Finance and Sales charts for the CEO role.	[DEV-3]	4.2
4.4	Department Dashboards	Build specific, detailed dashboard views for HR, Sales, and Warehouse managers using Chart.js/Apexcharts within Odoo.	[DEV-3]	4.2

Phase 5: AI Layer

Objective: Integrate Artificial Intelligence to assist ERP users.

ID	Task Name	Description	Assigned	Dependencies
5.1	LLM Gateway API	Securely connect the Odoo backend to an external LLM API (OpenAI/Claude) using a centralized authentication key.	[DEV-1]	None
5.2	Context Injection Engine	Build the backend system that safely queries the tenant's specific data to provide context to the LLM without leaking.	[DEV-1]	5.1
5.3	Anomaly Detection	Write background scheduled actions that detect outliers in inventory movements or sales trends and generate alert records.	[DEV-2]	None

5.4	NL to Domain Mapper	Build logic that translates natural language requests (e.g., "Show me sales > 1000") into safe Odoo ORM domain filters.	[DEV-2]	5.1
5.5	ERP Assistant Widget	Build a persistent, floating chat widget (OWL) across the Odoo interface for users to talk to the AI.	[DEV-3]	5.2
5.6	Smart Search UI	Overhaul the standard Odoo search bar to accept natural language, displaying results fed from Task 5.4.	[DEV-3]	5.4

Phase 6: Customer Portal

Objective: Allow the tenants' clients to interact with them securely.

ID	Task Name	Description	Assigned	Dependencies
6.1	Payment Gateways	Integrate regional payment gateways (Stripe, PayTabs, Tap) allowing customers to pay invoices directly via the portal.	[DEV-1]	2.6
6.2	Portal Access Rights	Strictly define XML record rules limiting what portal users can see (only their own Invoices, Orders, Tickets).	[DEV-2]	1.7
6.3	Support Ticketing	Implement Helpdesk/Ticketing backend logic enabling portal users to submit issues that route to the tenant's CRM.	[DEV-2]	None
6.4	Portal UI Redesign	Redesign the standard Odoo /my portal templates to match the NCollection premium, modern design language.	[DEV-3]	None
6.5	Knowledge Base UI	Build a self-service documentation layout for portal users to search for FAQs and guides.	[DEV-3]	None

Phase 7: Mobile Application

Objective: Deliver mobile accessibility for field workers and executives.

ID	Task Name	Description	Assigned	Dependencies
7.1	Mobile API Optimization	Secure and optimize Odoo XML-RPC/JSON-RPC endpoints for lightweight mobile consumption (pagination, stripped fields).	[DEV-1]	1.1
7.2	Push Notification Server	Setup Firebase Cloud Messaging (FCM) integration on the backend to trigger mobile alerts for approvals.	[DEV-1]	None
7.3	Offline Sync Logic	Implement conflict-resolution logic on the backend to handle data pushed from mobile devices that went offline.	[DEV-2]	7.1
7.4	Barcode Endpoints	Create hyper-optimized endpoints specifically for rapid warehouse movement barcode scanning.	[DEV-2]	None
7.5	Mobile App Development	Scaffold and develop the mobile application (React Native / Flutter) integrating with the Odoo API.	[DEV-3]	7.1

7.6	Mobile UI/UX Design	Design and implement mobile screens tailored specifically for Sales entry, Inventory scanning, and Manager approvals.	[DEV-3]	None
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Phase 8: Platform Services

Objective: Open the platform for enterprise integrations and monitoring.

ID	Task Name	Description	Assigned	Dependencies
8.1	Public REST API Layer	Build a standard REST API layer utilizing OAuth2 authentication, wrapping Odoo's internal RPC for public developer consumption.	[DEV-1]	None
8.2	Webhooks System	Implement an event-driven outgoing webhooks manager so tenants can push updates to external systems (e.g., Zapier).	[DEV-1]	None
8.3	System Monitoring	Setup Prometheus exporters and Grafana dashboards to monitor global platform health (CPU, DB connections, request latency).	[DEV-1]	1.1
8.4	Strict Audit Trail	Implement overriding logic on critical models (Accounting, CRM) to track exactly who changed what fields, when, and from where.	[DEV-2]	None
8.5	Developer SDK	Document the API endpoints and create a foundational Python and Node.js client SDK for third-party developers.	[DEV-2]	8.1
8.6	API Swagger/Redoc UI	Build a public-facing Swagger/Redoc UI webpage that automatically generates from the REST API routes.	[DEV-3]	8.1
8.7	Marketplace UI	Build an interface within the ERP where tenants can browse and install verified third-party integrations.	[DEV-3]	None